離散数学 課題06 22059 平間

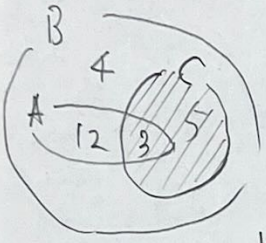
1

ex2.6

$A=\{1,2,3\}$

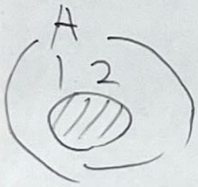
$B=\{1,2,3,4,5\}$

$C=\{3,5\}$

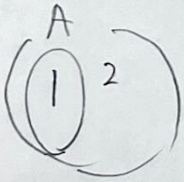


ex2.7

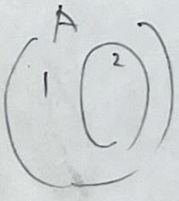
$A=\{1,2\}$



$\phi \subset A$



$\{1\} \subset A$



$\{2\} \subset A$

$\{1,2\} \subset A$

$P(A)=\{\phi,\{1\},\{2\},\{1,2\}\}$

$1\notin, 2\notin \phi$

$1\in, 2\notin \{1\}$

$1\notin, 2\in \{2\}$

$1\in, 2\in \{1,2\}$

$B=\{1,2,3\}$のべき集合

$P(B)=\{\phi,\{1\},\{2\},\{3\},\{1,2\},\{1,3\},\{2,3\},\{1,2,3\}\}$

ex2.8

$\{1,2,3,4\}$と$\{2,1,3,4\}$は相等

$\{x \mid x$は$2<x<10$である素数$\}$と

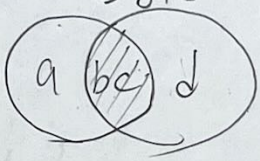
$\{x \mid x$は$1<x<8$である奇数$\}$も

相等

ex2.9

$\{a,b,c\} \cap \{b,c,d,e\}$

$\Rightarrow b,c$



$\{a,b,c\}$と$\{e,d\}$は

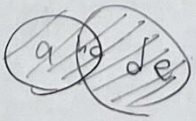
互いに素であり

$\{a,b,c\} \cap \{e,d\} = \phi$

ex2.10

$\{a,b,c\} \cup \{b,c,d,e\}$.

$\Rightarrow \{a,b,c,d,e\}$



ex2.11

$A \cup (A \cap B) = A$の証明

$A \cup (A \cap B) \subset A$と$A \cup (A \cap B) \supset A$

を示す

前者 任意xが$x \in (A \cup (A \cap B))$のとき

xはAまたは$A \cap B$の要素

いずれも$x \in A$がなりたつ。

$A \cup (A \cap B) \subset A$

後者 任意のxについて$x \in A$のとき、

$x \in A \cup (A \cap B)$である。よって$A \cup (A \cap B) \supset A$

以上のことから$A \cup (A \cap B) = A$

ex 2.12

$\{a,b,c\} - \{c,e,d,b\} \Rightarrow \{a\}$

ex2.13

$\mathbb{Z}$が全体集合で $\mathbb{Z} \supset \mathbb{N}$

$\mathbb{N}$の補集合$\mathbb{Z}-\mathbb{N}$は負の整数全体の集合

$\mathbb{Z}^-$である。

ex2.14

$L=\{1,2,3\}, R=\{a,b,c\}$

$L \times R = \{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c),(3,a),(3,b),(3,c)\}$

直積の要素は $3 \times 3 = 9$ 個である。

2

2.10

$N \subset N, N \subset Z, N \subset Q, N \subset R$

$Z \subset Z, Z \subset Q, Z \subset R$

$Q \subset Q, Q \subset R, R \subset R$

2.11

a) $\{\phi, \cancel{\{1,2,3\}}, \{1\}, \{2\}, \{3\}$
$\{1,2\}, \{2,3\}, \{1,3\}\}$

b) $\{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$

c) $\{\{x, y, z, u\}, \cancel{\{x\}}, \{y\}, \{z\},$
$\{u\}, \{x, y\}, \{x, z\}, \{x, u\}, \{y, z\}$
$\{y, u\}, \{z, u\}, \{x, y, z\}, \{y, z, u\},$
$\{x, z, u\}, \{x, y, u\}, \phi\}$

$|\{x, y, z, u\}| = 4 \neq 11$

$|P(\{x, y, z, u\})| = 2^4 = 16$

2.12

$\phi, \{そば\}, \{うどん\}, \{ラーメン\}, \{そば, うどん, ラーメン\},$
$\{そば, うどん\}, \{そば, ラーメン\}, \{うどん, ラーメン\}$

2.13

a, b

2.14 省

B⊂C, ⊂D

2.14

$A \cap (A \cup B) = A$ の証明

$A \cap (A \cup B) \subset A$

$x \in (A \cup (A \cap B))$ のとき x は A と A
x は A または $A \cap B$ の要素

$\rightarrow x \in A$

$\rightarrow A \cup (A \cap B) \subset A$

$A \cap (A \cup B) \supset A$

$x \in A$ のとき $x \in A \cup (A \cap B)$

よって $A \cup (A \cap B) \supset A$

よって $A \cup (A \cap B) = A$

2.16 $C = \{ア, イ, ウ, エ, オ\}$

$D = \{ア, カ, キ\}$

a) $C \cap D = \{ア\}$

b) $C - D = \{イ, ウ, エ, オ\}$ ~~カ, キ~~

c) $C - (C - D) = \{ア\}$ ~~カ, キ~~

2.17

a) $(A \cup B) - C$

b) ① $(A \cup B \cup C) - A \cup C$

$(B - A) - C$

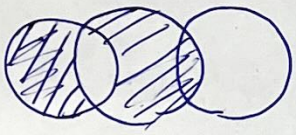
② $(A \cup B) - C$

$= \overline{A \cup B} - \overline{C}$

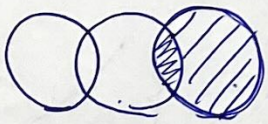
$= (\overline{A} \cap \overline{B}) - \overline{C}$

$C - (C - B)$

$(B - A) - C$

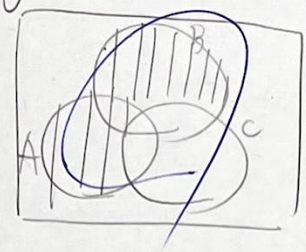


$C - (C - B)$

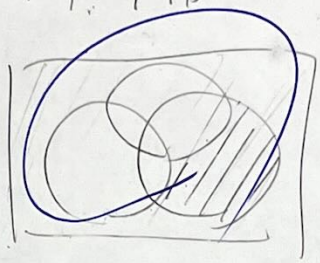


2.18

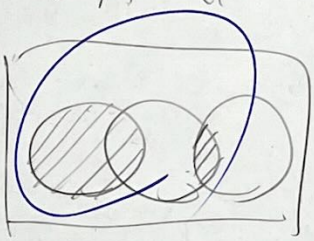
a) $(A \cup B) - C$



b) $A^c \cap B^c \cap C$



c) $A \cup (B \cap C)$



2.19

c) $A^2 \times A^3$

$A^2 = \{(0,0), (0,1), (1,0), (1,1)\}$

$A^3 = \{(0,0,0), (0,0,1), (0,1,0), (1,0,0), (1,1,0), (1,0,1), (0,1,1), (1,1,1)\}$

d) $A \times C = C \times A$

$C = A, \quad C = \phi$

2.20 $A = \{0,1\}$ $B = \{0,1,2\}$

a) $|A \times B| = \{(0,0), (0,1), (0,2), (1,0), (1,1), (1,2)\}$

b) $(A \times B) \cup (B \times A) =$

$B \times A = \{(0,0), (0,1), (1,0), (1,1), (2,0), (2,1)\}$

$\therefore \{(0,0), (0,1), (0,2), (2,0), (1,0), (1,2), (2,1)\}$

$(1,1)$

2.18

a) $A \cup B = \{A, \text{☆}, \text{⊥}, \text{⊟}\}$

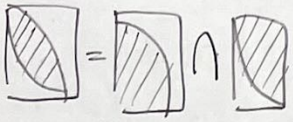
b) $A^c = \{\text{☆}, \text{⊥}, \text{⊟}\}$

c) $A \cap B \cap C = \{\text{☆}\}$ ϕ

d) $U - (A \cup B \cup C) = \{\text{☆}, \text{⊥}\}$

3

Ex1.3



Xの面積|X|とすると

$|A\cup B| = |A| + |B| - |A\cap B|$ …①

$|A\cup B| = 1$ で $|A| = |B| = \frac{\pi}{4}$

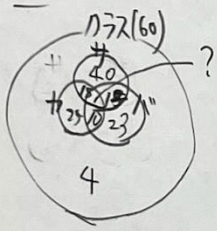
①より

$1 = |A\cup B| = \frac{\pi}{4} + \frac{\pi}{4} - |A\cap B|$

よって

$|A\cap B| = \frac{\pi}{2} - 1$

Ex1.4



ヰ = A, ヲ = B, ン = C

$|A| = 40, |B| = 25, |C| = 23$

$|A\cap B| = 15, |A\cap C| = 12, |B\cap C| = 10$

$\therefore 56 = 40 + 25 + 23 - 15 - 12 - 10 + |A\cap B\cap C|$

$\therefore |A\cap B\cap C| = 5$人

Q1.8

$|A\cup B|$?

$100 \div 3 = 33 = |A|$

$100 \div 4 = 25 = |B|$

$A\cap B = 12$, $|A\cap B| = 100 \div 12 = 8$

$|A\cup B| = |A| + |B| - |A\cap B|$ Q1.11

$= 33 + 25 - 8$

$= 58 - 8 = 5$

Q1.9

(1) $2^A = P(A)$

$= \{\phi, \{1\}, \{0\}, \{0,1\}\}$

$P(C) = \{\phi, \{\alpha\}, \{\beta\}, \{\gamma\}, \{\delta\},$
$\{\alpha,\beta\}, \{\alpha,\gamma\}, \{\alpha,\delta\},$
$\{\beta,\gamma\}, \{\beta,\delta\},$
$\{\gamma,\delta\}, \{\alpha,\beta,\gamma,\delta\}\}$

(2) $A\times C = \{(0,\alpha), (0,\beta), (0,\gamma), (0,\delta),$
$(1,\alpha), (1,\beta), (1,\gamma), (1,\delta)\}$

$B\times A = \{(a,0), (a,1), (b,0), (b,1), (c,0),$
$(c,1)\}$

$B^2 = \{a, b, c\} \times \{a, b, c\}$
$= \{(a,a), (a,b), (a,c), (b,a),$
$(b,b), (b,c), (c,a), (c,b), (c,c)\}$

$A^3 = \{(0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,0),$
$(1,0,1), (1,1,0), (1,1,1)\}$

Q1.10

$\mathcal{P} = \{\phi, \{a\}, \{b\}, \{c\}, \{d\}, \{a,b\}, \{a,c\}, \{a,d\},$
$\{b,c\}, \{b,d\}, \{c,d\}\}$

$C = \{\{a,b,c\}, \{a,b,d\}, \{b,c,d\}, \{a,c,d\}\}$

$P(A\cap B) = \{\phi, \{0\}, \{2\}, \{0,2\}\}$

Q1.12

$50 - 5 = 30 + 35 - |A\cap B|$

$|A\cap B| = 20$人

Q1.13

$100 = 80 + 45 + 40 - 25 - 20 + 8 - |A\cap B|$

$-|A\cap B| = 100 - 80 - 45 - 40 + 25 + 20 - 8$

$= 145 - 173 = -28$

$|A\cap B| = 28$人